

DSSATTools: A Python package for crop modeling with DSSAT

Diego Quintero¹, Manuel A. Andrade², and Vikalp Mishra³

¹ Wageningen University & Research, The Netherlands ² University of Nevada, Reno, NV, USA ³ The University of Alabama in Huntsville, Huntsville, AL, USA

DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

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Editor: [↗](#)

Submitted: 06 June 2026

Published: unpublished

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Summary

DSSATTools is an open-source Python package designed for crop modeling using the Decision Support System for Agrotechnology Transfer (DSSAT) ecosystem. It facilitates the end-to-end crop growth and development modeling pipeline natively in Python. Specifically, DSSATTools provides a programmatic wrapper around DSSAT's core simulation engine and features specialized object-oriented classes representing the distinct agronomic entities, soil profiles, environmental records, and management decisions considered during a simulation.

Statement of need

DSSAT ([Hoogenboom et al., 2019](#)) is a widely recognized software suite incorporating over 45 process-based models for the simulation of crop growth, development, and yield. The Cropping Systems Module (CSM) serves as the primary simulation engine of DSSAT, using multiple sub-modules to simulate individual components of the cropping system (e.g., hydrology, soil dynamics, and genetics). As a legacy Fortran program, the CSM executes via a command-line interface, requiring a rigid set of fixed-width parameters and a simulation configuration file known as the FILEX. While the standard desktop distribution for Windows features a Graphical User Interface (GUI) to assist users with file creation, execution, and plotting without text-command exposure, it limits scalability.

In modern agronomic applications, it is increasingly necessary to generate input files, orchestrate simulations, and process high-dimensional outputs programmatically. This requirement is vital for large-scale grid-based crop yield assessments, multi-parameter model calibrations, sensitivity analyses, and the deployment of DSSAT-CSM as a modular component in larger software architectures (such as web tools or web-based decision services). Although the raw CSM supports command-line execution, manually building or modifying input files across dozens of distinct variables is highly error-prone.

DSSATTools bridges this gap by acting as a native Python interface to the CSM. It handles the structural requirements of DSSAT inputs and outputs, allowing agricultural scientists, crop modelers, data scientists, and developers to deploy robust modeling pipelines entirely within the Python data-science ecosystem.

State of the field

Existing efforts to wrap crop simulation models have varied by language and framework. For R users, DssatR ([Alderman, 2020](#)) provides a functional interface following a traditional scripting paradigm, which remains popular among academic researchers but is less integrated into modern web services or production engineering environments.

39 Other crop models feature pure Python implementations. For instance, AquaCrop-OSPy (Kelly
40 & Foster, 2021) implements the AquaCrop model (Steduto et al., 2009) natively, and the pcse
41 (Python Crop Simulation Environment) package (Wit et al., 2019) encapsulates Wageningen
42 University models like WOFOST. Within the DSSAT landscape, previous automation projects
43 like PyDSSAT were released as collections of standalone execution scripts. However, PyDSSAT
44 lacked documentation, featured no capabilities for structured input file generation, and
45 development has been completely inactive since 2015.

46 DSSATTools overcomes these limitations by offering a unified, actively maintained, object-
47 oriented ecosystem that handles the complete DSSAT pipeline—from raw data ingestion to
48 post-simulation analysis.

49 Software design

50 Inspired by modern software design patterns observed in pcse, DSSATTools (Version 3) is
51 structured around a modular, object-oriented framework. The software modules mirror the
52 legacy layout of DSSAT input data files, rendering the architecture intuitive to anyone familiar
53 with the DSSAT GUI tools.

54 The package is organized into four core modules corresponding to primary DSSAT components:

- 55 ■ **DSSATTools.soil**: Manages soil profiles (SoilProfile class) and translates structured
56 layers into the required .SOL format.
- 57 ■ **DSSATTools.weather**: Encapsulates environmental and meteorological observations into
58 a WeatherStation class to compile .WTH files.
- 59 ■ **DSSATTools.crop**: Controls genetic, cultivar, and ecotype parameters across 19 supported
60 crop types (e.g., the Maize class) by reading and modifying .CUL, .ECO, and .SPE files.
- 61 ■ **DSSATTools.filex**: Models individual experimental treatments and simulation
62 configurations represented within the DSSAT FILEX.

63 Individual operational sequences within the filex module are mapped directly to agronomic
64 management events. Single-instance tasks are managed via explicit definitions such as
65 Planting or FertilizerEvent, while repeated iterations across a season are collected into
66 structured containers like Fertilizer. Global operational directives are configured using the
67 SimulationControls class.

68 Simulations are executed by creating an instance of the DSSAT environment, which handles the
69 secure setup and tear-down of low-level scratch directories, protecting execution states during
70 parallel execution. Isolated single-treatment runs are dispatched via the run_treatment()
71 method, which outputs structured pandas.DataFrame execution tables and native text strings
72 representing standard outputs like PlantGro and OVERVIEW.

73 Research impact statement

74 DSSATTools has been integrated into a variety of peer-reviewed research areas involving digital
75 agriculture, climate resilience, and automated workflows. It has facilitated high-throughput
76 automated calibration pipelines across expansive geographic boundaries (Quintero et al., 2024,
77 2025) and enabled evaluations of climate-resilient crop ideotypes under variable management
78 practices (Srivastava, 2026). Additionally, the package has served as a critical execution engine
79 for modern decision support systems, such as coupling biophysical crop simulations with Large
80 Language Models (LLMs) to power conversational AI agents for agricultural decision-making
81 (Kpodo et al., 2026), and conducting regional risk management and intervention assessments
82 (Quintero et al., 2023).

83 AI usage disclosure

84 During the preparation of this manuscript, a generative AI model (Gemini by Google) was
 85 utilized strictly to refine technical terminology, improve narrative flow, and audit the Markdown
 86 structural formatting for compliance with journal layout standards. The authors maintain
 87 ultimate accountability for the intellectual content and factual accuracy of this paper.

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